

ay: _____

Solution
(Fall-23)

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Qno 1

hash function: $h(k) = k \bmod 11$

keys: 19, 22, 33, 44, 55, 80

quadratic probing
→ formula:

$$h(k) = (h(k) + i^2) \bmod 11$$

Solution :

Key	hash function	index
19	$19 \% 11$	8
22	$22 \% 11$	0
33	$33 \% 11$	0
44	$44 \% 11$	0
55	$55 \% 11$	0
80	$80 \% 11$	3

0	22
1	33
2	
3	80
4	44
5	
6	
7	
8	19
9	55
10	

~~Proof:~~ using : $(h(k) + i^2) \bmod 11$

33: $0 + 1^2 \% 11 = 1$ 33 go to index 1

44: $0 + 1^2 \% 11 = 1$
 $0 + 2^2 \% 11 = 4$ 44 go to index 4

55:

$$(0+1^2) \% 11 : 1 \rightarrow \text{occupied}$$

$$(0+2^2) \% 11 : 4 \rightarrow //$$

$$(0+3^2) \% 11 : 9 \rightarrow 55 \text{ will go to 9 index}$$

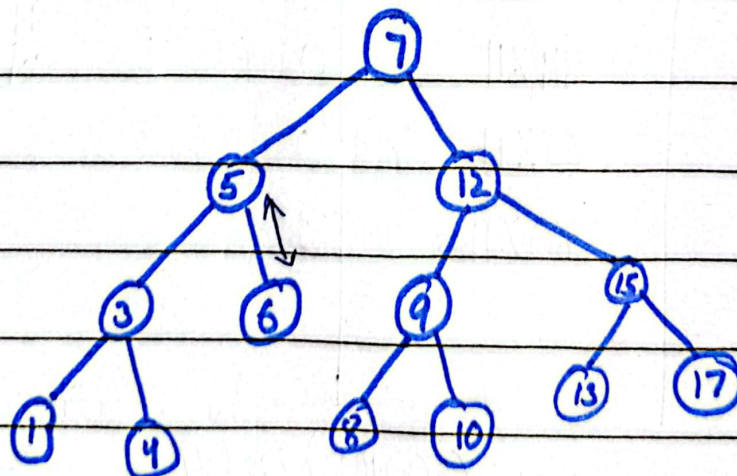
Qno 2

code on merge sorting:

Qno 3

(a)

Binary search tree:



Remove : 5 and 12

step by step:

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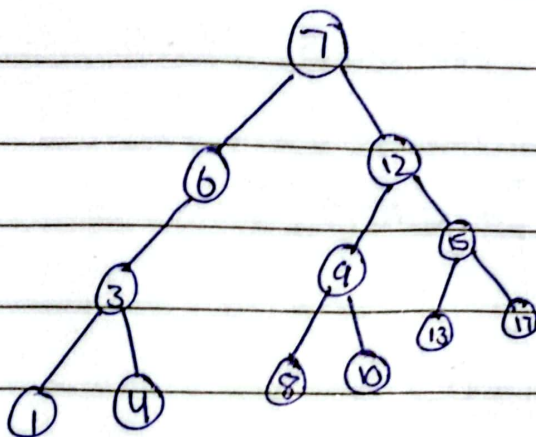
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Removal of 5

→ we will check smallest value in the right subtree

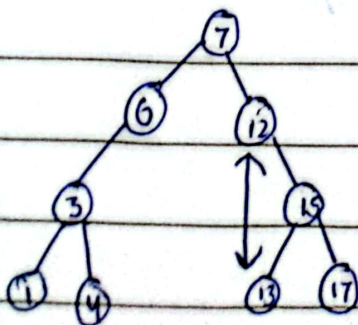
So the tree will become

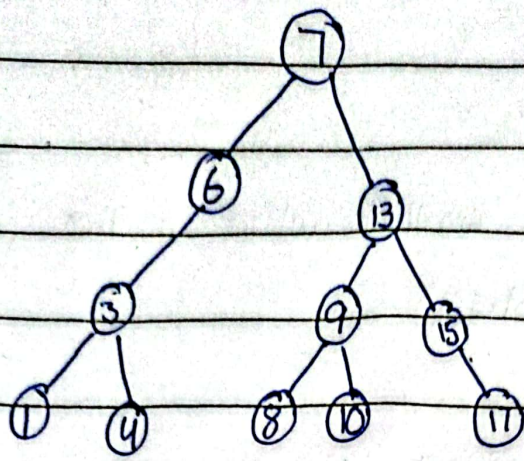
∴ we will move 6 to 5 position and remove 5



No remove 12

we will check the smallest value in right subtree of 12 that is 13
we replace 13 with 12 and remove 12
tree will become.





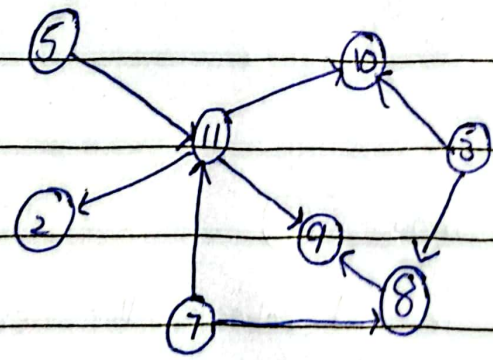
final tree after deletion:

(b)

code for checking identical

Question: 4

Graph



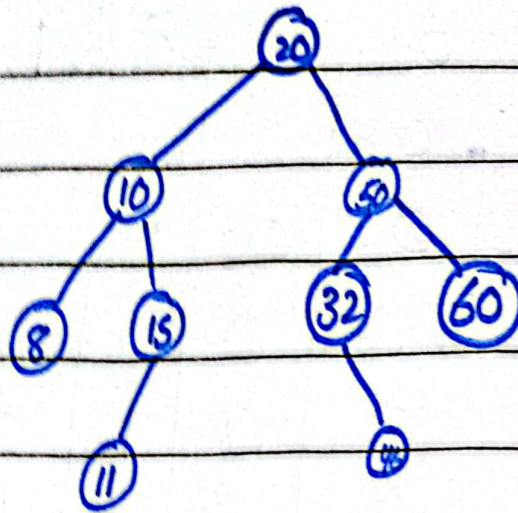
Draw adjacency matrix;

Adjancey matrix:

	2	3	5	7	8	9	10	11
2	0	0	0	0	0	0	0	0
3	0	0	0	0	1	0	1	0
5	0	0	0	0	0	0	0	1
7	0	0	0	0	1	0	0	1
8	0	0	0	0	0	1	0	0
9	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0
11	1	0	0	0	0	1	1	0

Adjancey list:2 $\rightarrow$ Null3 $\rightarrow$ [8] $\rightarrow$ [10] /5 $\rightarrow$ [11] /7 $\rightarrow$ [8] $\rightarrow$ [11] /8 $\rightarrow$ [9] /9 $\rightarrow$ Null10 $\rightarrow$ Null11 $\rightarrow$ [2] $\rightarrow$ [9] $\rightarrow$ [10] /

Question 5

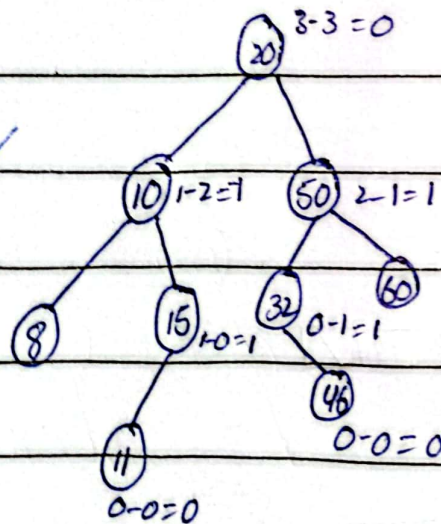


Solution

is initial tree is AVL?

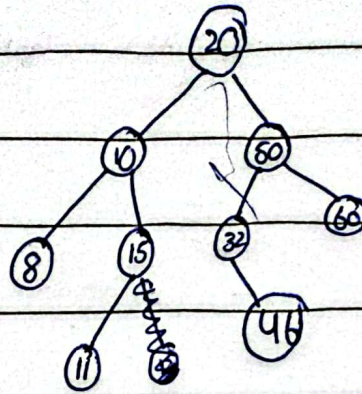
by check balance factor

we travel
to the
depth of
each
side

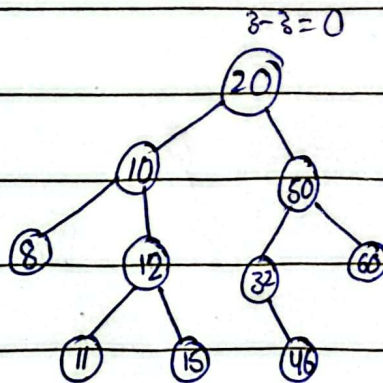
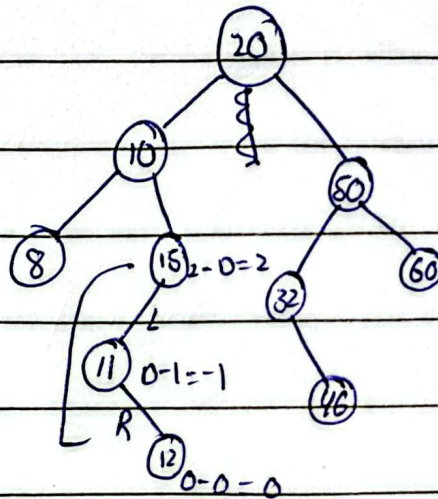


Yes, initial tree is AVL.

a) Insert 12 and 16 one by one.



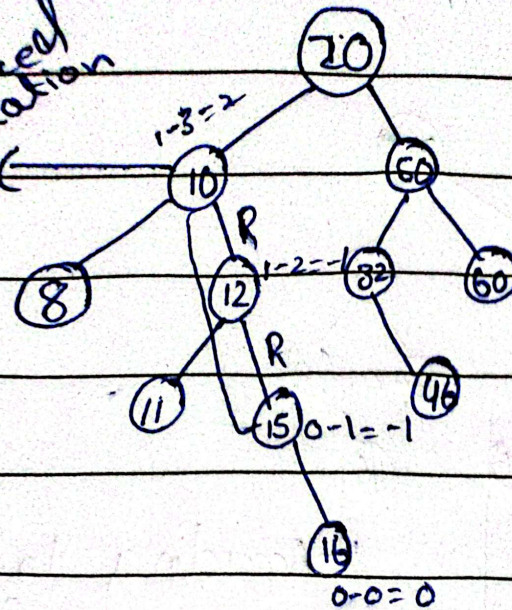
Solution



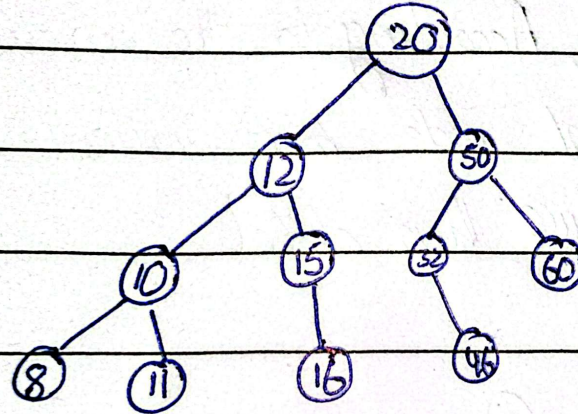
balanced

Now insert 16 :

Not balanced
RR rotation



$$3-3=0$$

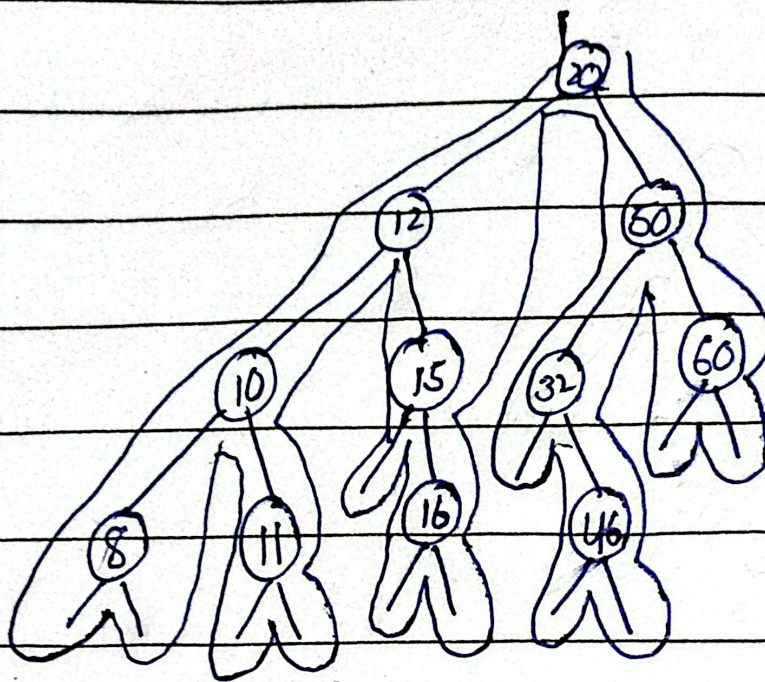


balanced.

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(b)

Inorder traversal: left $\rightarrow$ root $\rightarrow$ Right



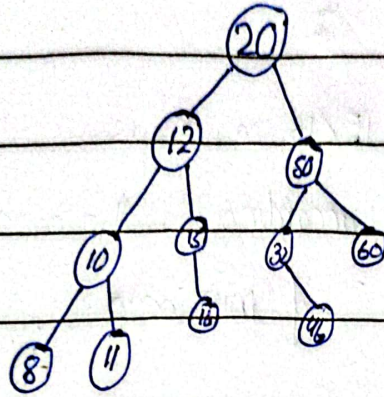
Inorder traversal:

8, 10, 11, 12, 15, 16, 20, 32, 46, 50, 60

Ans.

c) Min heap (heapify)

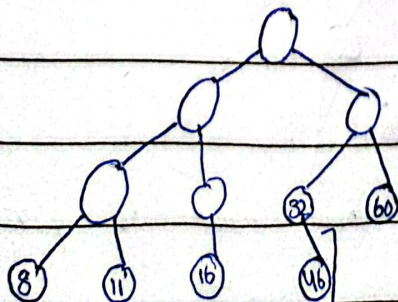
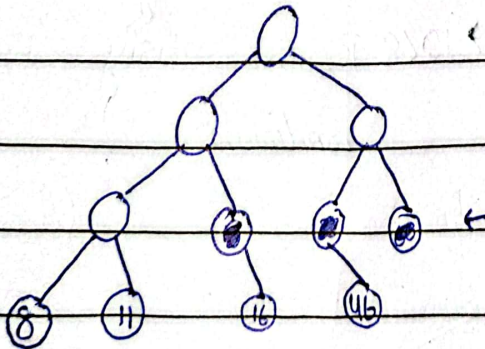
Given tree:



level by level filling:

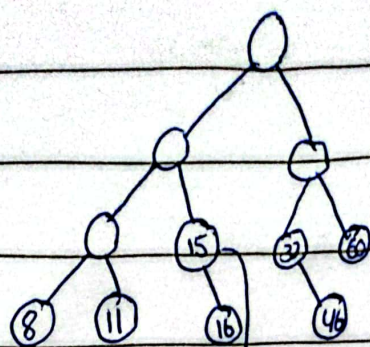
firstly fill the leaf nodes.

Step:



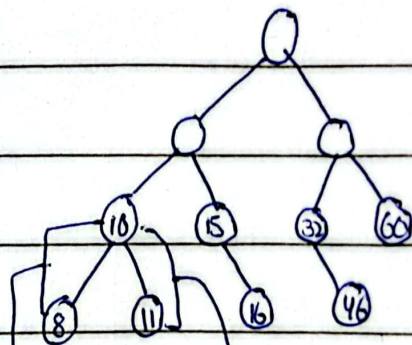
According to the condition of min heap.

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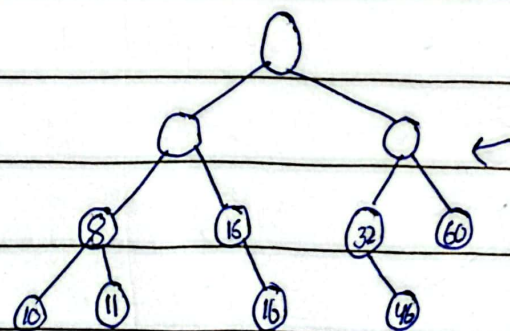
$15 < 16$ so it is

according to the condition
of min heap.



$10 < 11$ is according to condition.

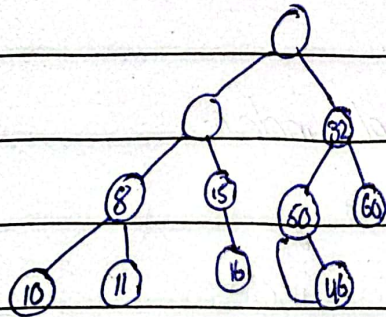
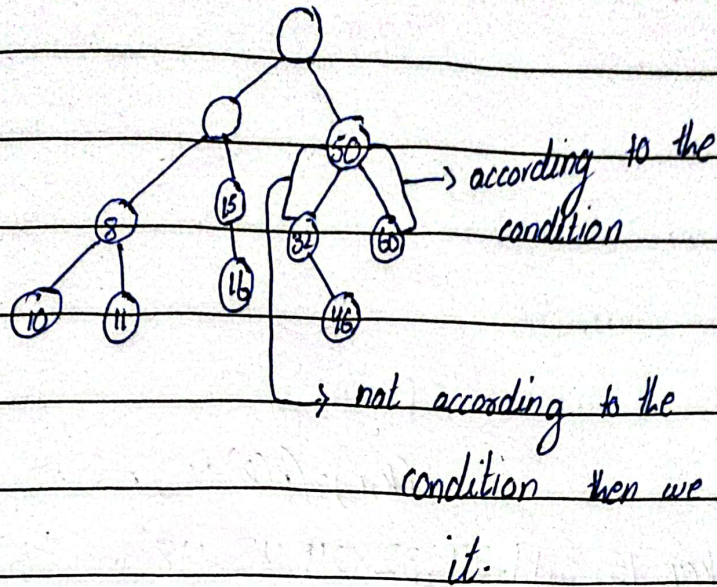
$10 < 8$ is not satisfy the ~~condition~~
condition.



now move to upper
level

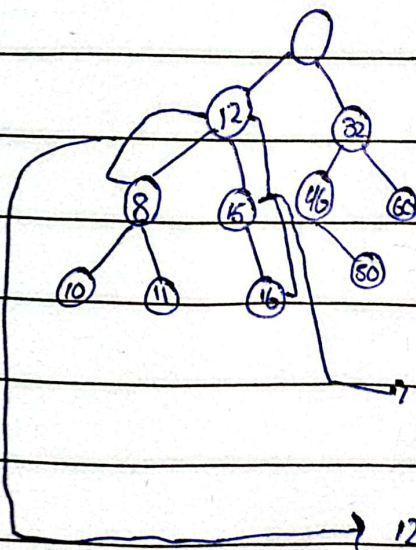
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$50 < 46$ false

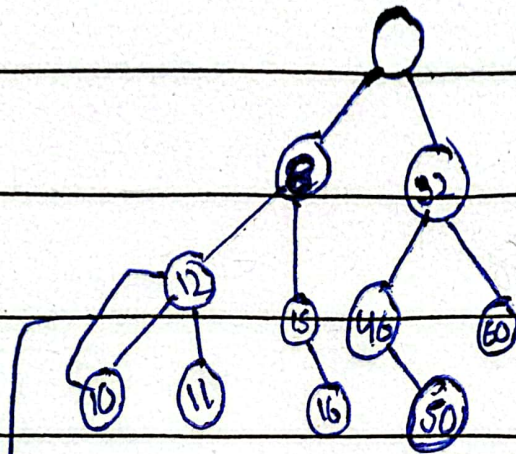
condition ~~does not~~ don't meet. swap it.



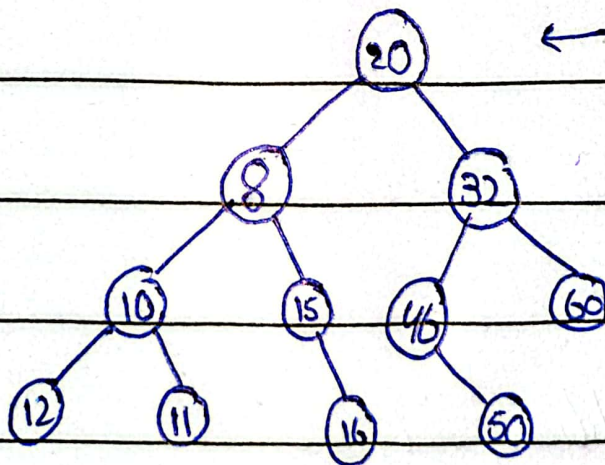
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Algorithm

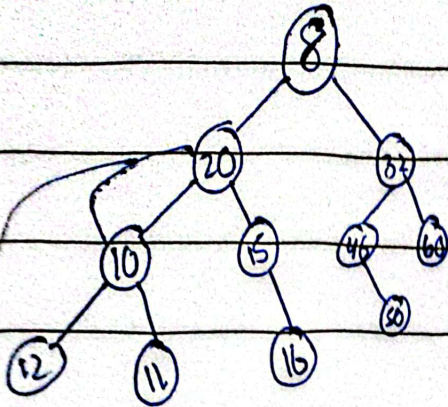
Date: _____



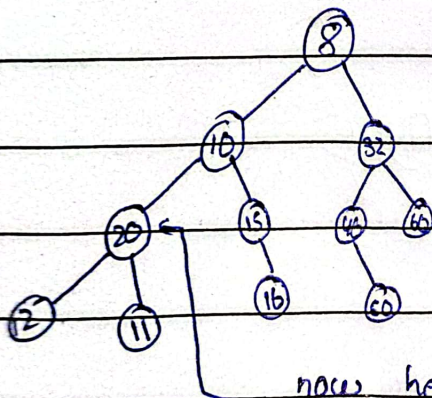
, $12 < 10$ false condition of min heap
not meet. so swap them



move to upper level



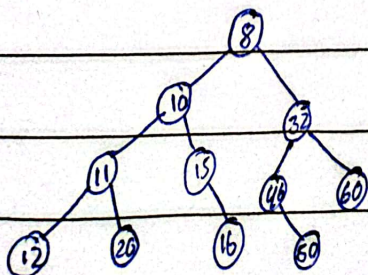
→ 20 < 10 false condition of min heap
not satisfied. swap it.



now here

20 < 11 false
20 < 12 false } both are false
so we chose
the minimum
value for this
point.

so we swap 20 with 11.



final tree which
is according to
Min heap
conditions.